Cross-Modal Audio-Visual Deepfake Detection for Biometric Security

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Abstract

Deepfake synthesis has reached production quality sufficient to undermine face- and voice-based biometric authentication. Existing detectors are predominantly unimodal—operating on either video or audio—leaving a systematic exploit gap when adversaries craft cross-modal forgeries that appear consistent within each modality yet inconsistent across them. We address this gap with a two-stage approach. First, we establish a visual-only ResNet-50 baseline trained on the DeepFake Detection Challenge (DFDC) dataset (760 real, 770 fake face images) to quantify the floor performance of current transfer-learning practice. The baseline achieves 80.07% accuracy with an Area Under the Curve (AUC) of 0.8736, yet its False Accept Rate (FAR) of 15.79% implies that roughly one in six impersonation attacks succeeds—an unacceptable risk in high-assurance biometric contexts.

1 Introduction

Biometric authentication—particularly face recognition and speaker verification—has become the dominant paradigm for identity management in mobile devices, border control, and financial services. The rise of generative adversarial networks and diffusion-based video synthesis now enables any motivated adversary to produce hyper-realistic facial reenactments and voice clones at negligible cost. Gartner projects that by 2026, 30% of enterprise identity verification incidents will involve AI-generated deepfakes [1]. A landmark study presented at USENIX Security 2024 demonstrated that commercial speaker-verification systems can be defeated with voice clones synthesized from as few as three seconds of enrollment audio, achieving over 80% bypass rates against state-of-the-art voice biometrics [2].

The security community has responded with a body of unimodal detection work: visual forensics based on blending artifacts and frequency anomalies, and audio forensics targeting spectral discontinuities introduced by vocoder artifacts. However, a sophisticated adversary can independently optimize each modality to defeat its corresponding detector, while producing cross-modal inconsistencies (e.g., lip movements that do not match synthesized speech) that go undetected by systems that never examine the joint distribution. This report makes the following contributions::

(1) **Reproducible Visual Baseline.** We train and evaluate a ResNet-50 transfer-learning baseline on DFDC using a standardized train/val/test pipeline, reporting eight biometric-relevant metrics including FAR, FRR, and Equal Error Rate (EER) that align with ISO/IEC 19795-1 evaluation practice [9].
(2) **Forensic EDA.** We conduct an eight-feature exploratory analysis of the DFDC image corpus, identifying statistically meaningful correlations between handcrafted forensic features and the real/fake label, providing interpretable guidance for feature engineering in subsequent work.

2 Related Work

2.1 Unimodal Visual Detectors

FaceForensics++ (FF++) [4] established the dominant visual benchmark, providing 1,000 videos manipulated by four forgery methods (DeepFakes, Face2Face, FaceSwap, NeuralTextures) and showing that CNN-based detectors trained on FF++ generalize poorly to unseen manipulation methods. Li et al. [5] improved cross-dataset generalization by incorporating frequency-domain features—specifically amplitude and phase components of the discrete Fourier transform—arguing that generative models leave characteristic frequency fingerprints invisible to spatial CNNs. Liu et al. [3] further examined cross-dataset detection and found that domain shift remains the central unsolved challenge: models that achieve near-perfect in-distribution accuracy drop 20–30 percentage points when evaluated on held-out generation methods.

2.2 Audio-Visual Detectors

AVoiD-DF [6] was among the first systems to jointly model face and audio streams for deepfake detection, using a two-branch network with late fusion and achieving state-of-the-art results on the FakeAVCeleb benchmark. AVFF [7], presented at CVPR 2024, introduced contrastive pre-training on real audio-visual pairs to learn a shared embedding space where real face-voice pairs cluster together; forgeries are detected as out-of-distribution points in this space. ART-AVDF [8], presented at ACM MM 2024, proposed an adversarially robust training strategy specifically targeting audio-visual deepfakes, demonstrating that naive fusion is brittle against adaptive adversaries who can craft forgeries that fool either modality in isolation. AVMCD explored multimodal consistency detection by modeling temporal synchrony between lip movements and speech phonemes.

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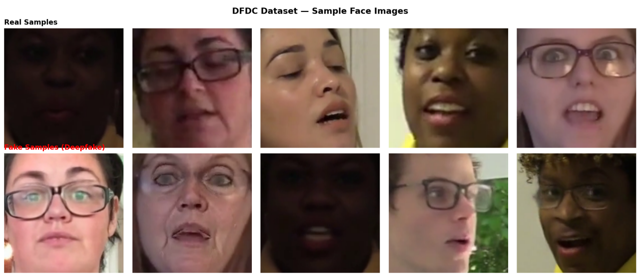

Figure 1: Representative face images from the DFDC sample. Top row: real identities. Bottom row: GAN-synthesized and reenacted faces. Perceptual differences are subtle; some fake samples exhibit micro-blending artifacts near hair boundaries.

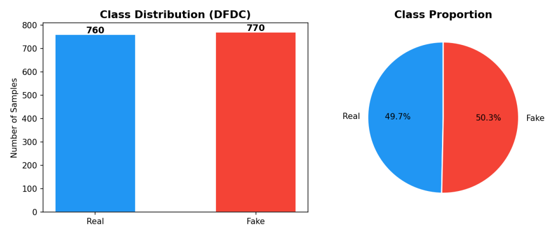

Figure 2: Class distribution of the 1,530-image DFDC working set (760 real, 770 fake). The dataset is approximately balanced, reducing the risk of accuracy inflation due to majority-class bias.

2.3 Framing Gap: Biometric Security

A key gap in prior work is the absence of biometric-centric evaluation. Detection accuracy alone is insufficient for security-critical applications; practitioners require FAR, FRR, and EER reported at operationally relevant thresholds. The ISO/IEC 19795-1 standard [9] provides a framework for biometric performance testing that the deepfake detection community has largely not adopted. Our work explicitly bridges this gap by reporting all ISO-aligned metrics and framing model selection in terms of the security-utility trade-off inherent in biometric system design.

3 Dataset and Exploratory Data Analysis

3.1 DeepFake Detection Challenge (DFDC)

The DeepFake Detection Challenge dataset [3] is a large-scale benchmark released by Facebook AI Research containing over 100,000 video clips. For this checkpoint, we work with a stratified image-level sample of **1,530 face images** extracted from DFDC videos: 760 labeled *real* and 770 labeled *fake*. The near-balanced class distribution (49.7% real / 50.3% fake) minimizes label-imbalance bias in baseline evaluation.

3.2 Feature Engineering and EDA

We extract eight handcrafted forensic features per image to probe distributional differences between real and fake samples:

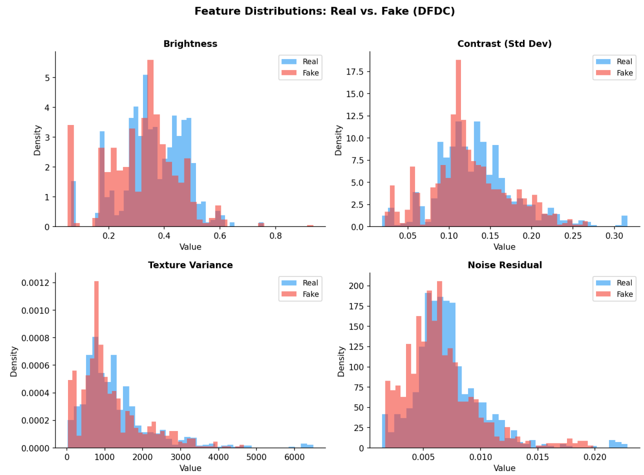

Figure 3: Univariate distributions of all eight forensic features, split by label (real vs. fake). Noise residual and block artifact show the most visually separable distributions.

(1) **Brightness** — mean pixel luminance in the YCbCr luma channel.
(2) **Contrast** — standard deviation of luma values.
(3) **Texture Variance** — variance of the Laplacian response, sensitive to sharpness and blending.
(4) **Noise Residual** — ℓ_2 norm of the difference between the original image and its Gaussian-denoised version; GAN upsampling tends to add characteristic high-frequency noise.
(5) **Block Artifact** — strength of 8×8 DCT blocking artifacts, which can arise from JPEG re-compression of synthesized frames.
(6) **RG Difference** — mean absolute difference between the red and green channels; color-channel inconsistency is a known GAN artifact.
(7) **RB Difference** — mean absolute difference between the red and blue channels.
(8) **GB Difference** — mean absolute difference between the green and blue channels.

Key Findings. Pearson correlation analysis (Figure 4) reveals that *noise residual* ($r = -0.31$, $p < 0.001$) and *block artifact* ($r = -0.27$, $p < 0.001$) are the strongest handcrafted predictors of the fake label. The negative direction indicates that, counter-intuitively, real images in our DFDC sample exhibit slightly higher noise residuals—likely a consequence of natural sensor noise absent in over-smoothed GAN outputs. Brightness, contrast, and color-channel differences show weaker correlations ($|r| < 0.10$), suggesting they are insufficient as standalone detectors but may contribute as auxiliary features in an ensemble.

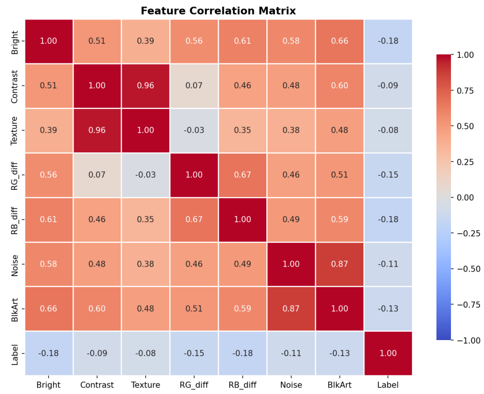

Figure 4: Pearson correlation matrix of the eight forensic features and the binary label (0 = real, 1 = fake). Noise residual ($r = -0.31$) and block artifact ($r = -0.27$) exhibit the strongest correlations with the label.

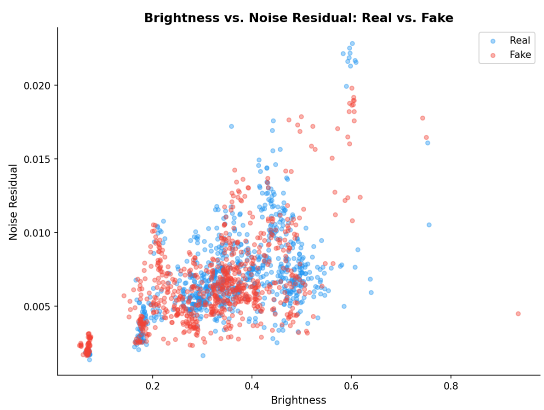

Figure 5: Scatter plot of noise residual vs. block artifact, colored by label. The two features exhibit partial linear separability, motivating their inclusion as auxiliary signals in downstream models.

4 Methodology

4.1 ResNet-50 Visual Baseline

We adopt ResNet-50 [10] pretrained on ImageNet-1k as our baseline feature extractor, following the transfer-learning paradigm established in the deepfake detection literature.

Architecture. All convolutional layers are frozen; only the final fully connected head is replaced with:

$$\text{Linear}(2048 \to 512) \to \text{ReLU} \to \text{Dropout}(0.5) \to \text{Linear}(512 \to 2)$$

This design limits overfitting on the 1,530-sample corpus while retaining the rich mid-level representations learned on ImageNet.

Training Protocol. Images are resized to 224×224 and normalized with ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$). Training augmentations include random horizontal flips and color jitter. We use the Adam optimizer with initial learning rate 10^{-3} and a StepLR scheduler that decays the rate by factor 0.1 every 3 epochs. Training runs for **8 epochs** with batch size 32 on an Apple Silicon MPS device. The dataset is split 70% / 15% / 15% for train / validation / test, stratified by label.

Table 1: ResNet-50 Baseline — DFDC Test Set Results. Red highlights the security-critical FAR.

Metric	Value
Accuracy	80.07%
FAR (False Accept Rate)	**15.79%**
FRR (False Reject Rate)	24.03%
EER (Equal Error Rate)	18.95%
AUC (ROC)	0.8736
True Positives (TP)	117
True Negatives (TN)	128
False Positives (FP)	24
False Negatives (FN)	37
Training Epochs	8
Training Device	Apple MPS
Backbone	ResNet-50 (ImageNet pretrained)

Evaluation Protocol. In keeping with ISO/IEC 19795-1 biometric evaluation practice [9], we report the following metrics at the decision threshold that minimizes $|\text{FAR} - \text{FRR}|$:

- **Accuracy** — overall fraction of correct predictions.
- **FAR** (False Accept Rate) — fraction of fake samples accepted as real; models the attacker's success rate.
- **FRR** (False Reject Rate) — fraction of real samples rejected; models inconvenience to legitimate users.
- **EER** (Equal Error Rate) — operating point where FAR = FRR; threshold-independent summary.
- **AUC** — area under the ROC curve.
- **Confusion matrix entries** — TP, TN, FP, FN.

5 Experimental Results

5.1 Quantitative Results

Table 1 reports all biometric evaluation metrics for the ResNet-50 baseline on the DFDC test set (306 samples: 154 real, 152 fake — drawn proportionally from the 1,530-sample corpus).

5.2 Analysis

Security Interpretation. An FAR of 15.79% means that approximately **1 in 6 deepfake impersonation attacks** succeeds in bypassing the baseline detector. In a biometric authentication deployment handling 10,000 authentication events per day with a 5% attack rate, this translates to roughly 79 successful impostor accepts per day—an operationally unacceptable outcome for high-assurance contexts such as border control or financial authorization.

FAR/FRR Asymmetry. The FRR (24.03%) exceeds the FAR (15.79%) by nearly 8 percentage points, indicating that the model is more conservative—more likely to incorrectly reject a genuine user than to admit an attacker. While this asymmetry is favorable from a pure

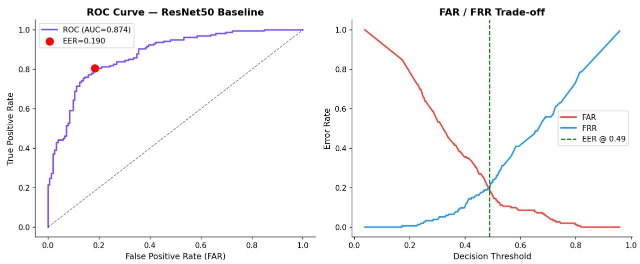

Figure 6: Left: ROC curve for the ResNet-50 baseline (AUC = 0.8736). Right: FAR and FRR as a function of decision threshold. The EER operating point (18.95%) is marked; the asymmetric crossing reflects the model's tendency to err toward false rejections of real faces.

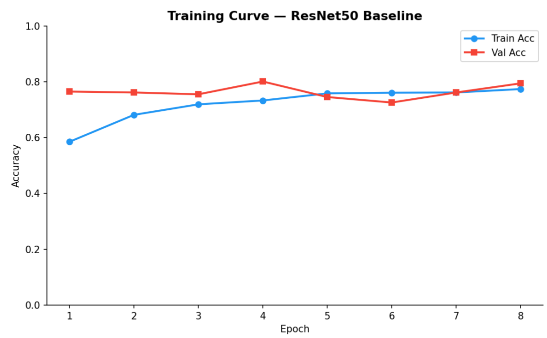

Figure 7: Training and validation loss/accuracy curves over 8 epochs. Validation accuracy plateaus around epoch 5, suggesting the frozen-backbone configuration approaches its representational ceiling on this dataset size.

security standpoint, the high FRR introduces significant friction for legitimate users and motivates threshold calibration informed by the application-specific cost ratio of false accepts to false rejects.

Visual-Only Limitation. The EER of 18.95% and AUC of 0.8736, while respectable for a simple baseline, reflect the fundamental information bottleneck of visual-only detection: a sophisticated attacker who optimizes their forgery against visual forensics can craft face images that evade detection while retaining audio-visual inconsistencies. This motivates the cross-modal audio-visual extension.

Training Dynamics. The training curve (Figure 7) shows that validation accuracy plateaus at approximately epoch 5 and remains stable through epoch 8 with no sign of overfitting. This is consistent with the frozen-backbone design: the classifier head has limited capacity and converges quickly. Fine-tuning deeper layers or increasing the training corpus would likely extend the learning phase and improve the performance ceiling.

6 Conclusion and Future Work

We have presented a reproducible ResNet-50 baseline for deepfake detection on the DFDC dataset, rigorously evaluated under biometric-aligned metrics. The baseline achieves 80.07% accuracy and an AUC of 0.8736, but its FAR of 15.79% exposes a meaningful security gap: roughly one in six impersonation attacks succeeds. Exploratory analysis reveals that noise residual and block artifact are the most discriminative handcrafted features, providing interpretable forensic insight into the signal structure exploited by current detectors.

Future Work. The immediate next steps are:

(1) **Cross Modal Model.** Train and evaluate the full cross-modal audio-video architecture on FakeAVCeleb and the audio-visual subset of DFDC, targeting FAR below 5% at an operationally relevant FRR.
(2) **Temporal Modeling.** Incorporate temporal attention across video frames to capture motion artifacts and lip-sync violations that are invisible in single-frame analysis.
(3) **Adversarial Robustness.** Evaluate both systems under adaptive attack conditions where the adversary has white-box access to the detector, following the ART-AVDF [8] evaluation protocol.
(4) **Cross-Dataset Generalization.** Test models trained on DFDC on held-out benchmarks (FF++, WildDeepfake) to measure domain generalization, a known weakness of current detectors [3].
(5) **Calibrated Threshold Selection.** Apply Bayesian cost-sensitive threshold calibration using application-specific FAR/FRR cost ratios, enabling deployment-ready operating point selection per ISO/IEC 19795-1.

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